

LAB EXPERIMENT RECORD

SET 7: Customer Demographics Analysis

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	SATURDAY

SET 7: Product Inventory Management Aim

To visualize customer demographic data using a bar chart, pie chart, and data table in R, and to build an interactive dashboard using Tableau..

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

PROGRAM 1: Bar Chart (Distribution of Customer Ages)

Aim: To visualize the distribution of customer ages using a bar chart with appropriate labels and title.

```
graphics.off()
```

```
customer_id <- c("1", "2", "3")
```

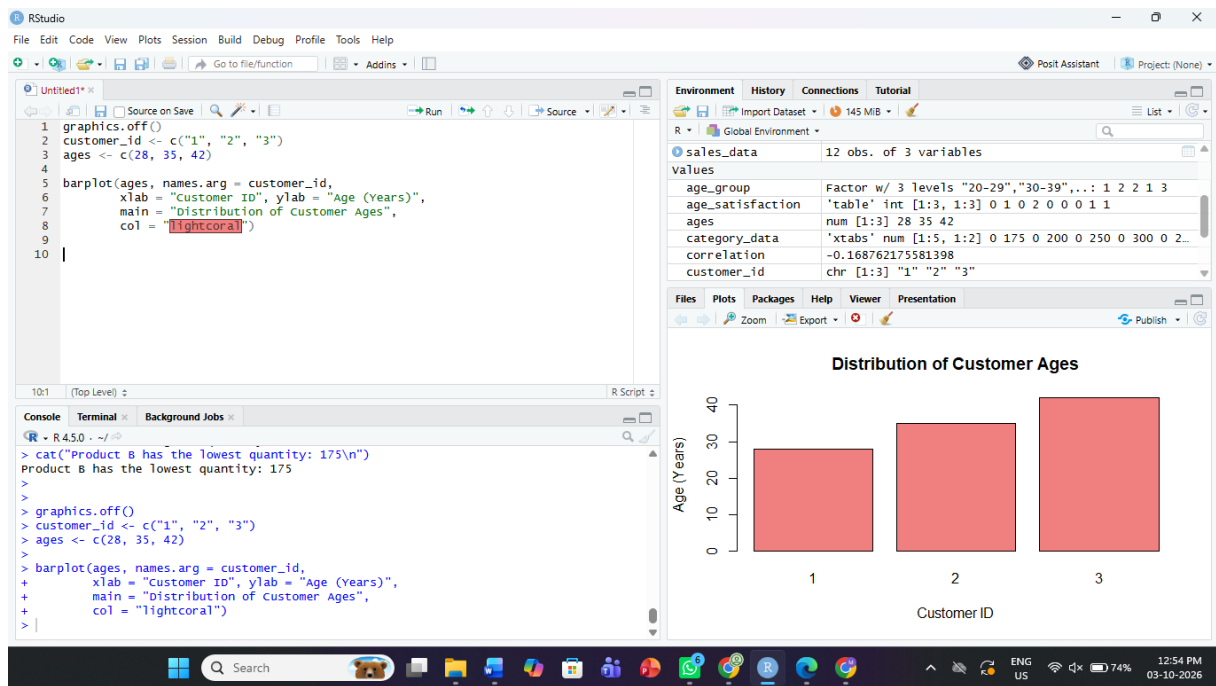
```
ages <- c(28, 35, 42)
```

```
barplot(ages, names.arg = customer_id,
```

```
  xlab = "Customer ID", ylab = "Age (Years)",
```

```
  main = "Distribution of Customer Ages",
```

```
  col = "lightcoral")
```



Observation: Customer 3 represents the oldest demographic at 42 years, while Customer 1 represents the youngest demographic at 28 years.

PROGRAM 2: Pie Chart (Gender Distribution)

Aim: To generate a pie chart representing the proportion of customers across genders.

```
graphics.off()
```

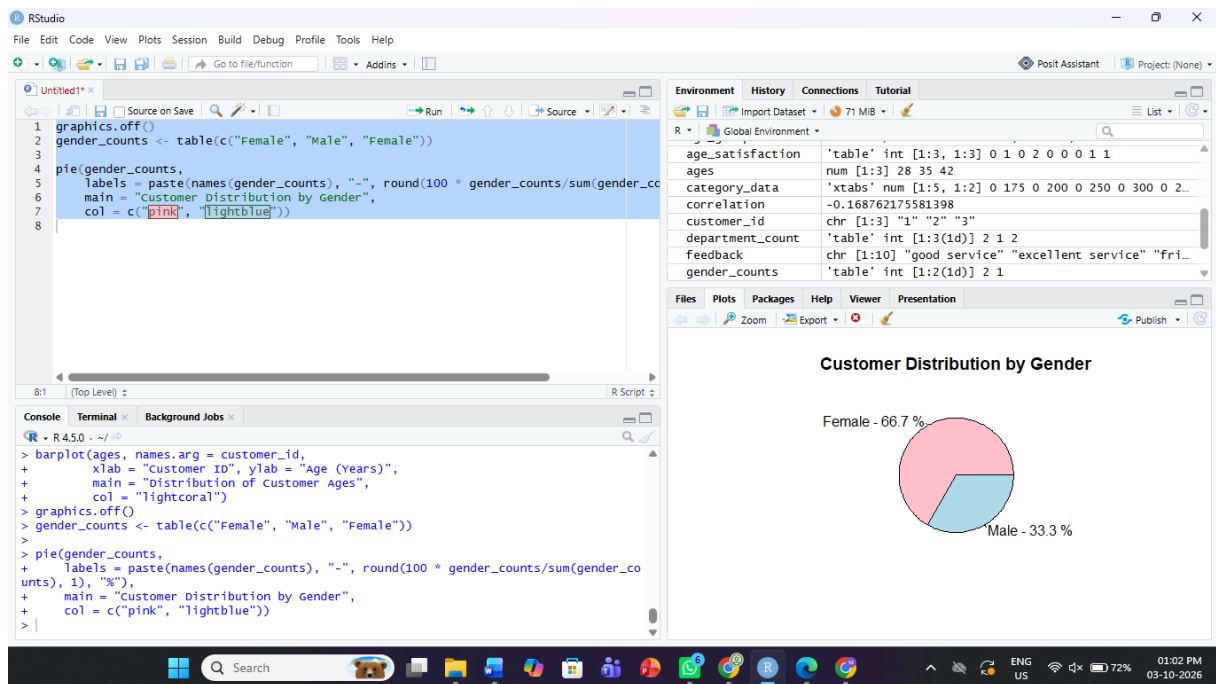
```
gender_counts <- table(c("Female", "Male", "Female"))
```

```
pie(gender_counts,
```

```
  labels = paste(names(gender_counts), "-", round(100 *
gender_counts/sum(gender_counts), 1), "%"),
```

```
  main = "Customer Distribution by Gender",
```

```
  col = c("pink", "lightblue"))
```



Observation: Females make up the majority of the sample pool at 66.7%, whereas males constitute 33.3%.

PROGRAM 3: Demographic Information Table

Aim: To build a structured summary table showing comprehensive demographic profiles.

```
customer_data <- data.frame(
```

```
  CustomerID = c(1, 2, 3),
```

```
  Age = c(28, 35, 42),
```

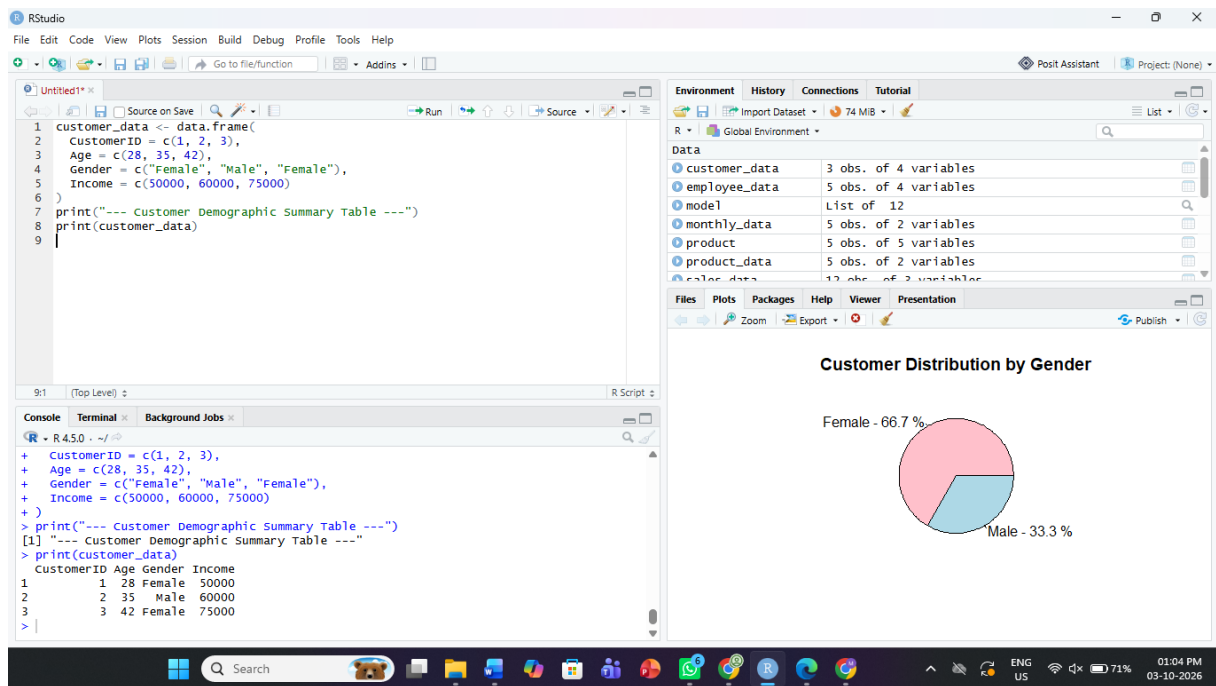
```
  Gender = c("Female", "Male", "Female"),
```

```
  Income = c(50000, 60000, 75000)
```

```
)
```

```
print("--- Customer Demographic Summary Table ---")
```

```
print(customer_data)
```



Observation: As age increases among the sampled individuals, there is a corresponding upward trend in income level.

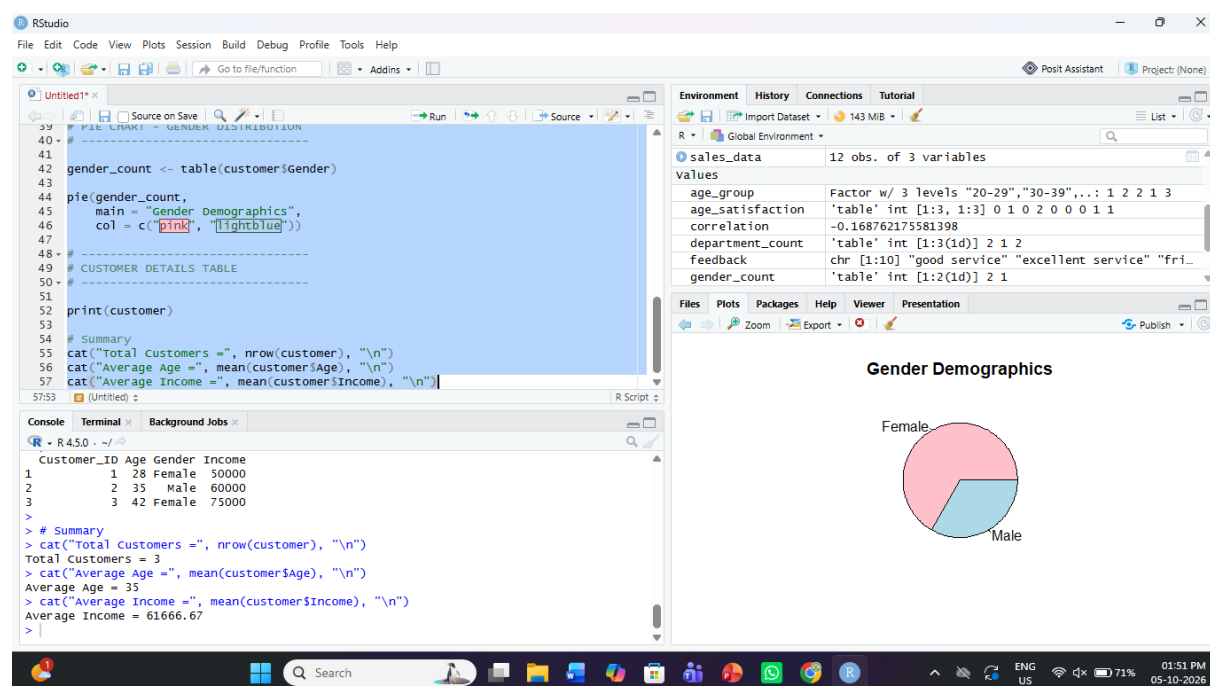
PROGRAM 4: Interactive Demographics Dashboard in Tableau

Aim: Develop a Tableau dashboard combining the bar chart, pie chart, and table for interactive exploration.

Steps in Tableau:

1. Save the customer dataset as a CSV file (Customer_Demographics.csv). Open Tableau and connect to the file using Connect > Text file.
2. Sheet 1 (Age Bar Chart): Drag Customer ID to Columns and Age to Rows (SUM). Set chart type to Bar. Add Age to labels. Title the sheet "Age Distribution".
3. Sheet 2 (Gender Pie Chart): Drag Gender to Color on the Marks card. Change mark type to Pie. Drag Customer ID (Count) to Angle and Label. Title the sheet "Gender Demographics".
4. Sheet 3 (Demographics Table): Drag Customer ID, Age, Gender, and Income into Rows to create a structured text table. Title the sheet "Customer Details Matrix".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size layout to Automatic, and drag all three sheets onto the layout workspace.
6. Click Use as Filter on the charts to activate full cross-filtering capability across visual components.

OUTPUT:



Observation: Clicking on a specific gender partition in the pie chart seamlessly isolates and highlights the corresponding records inside both the age bar chart and data details view.

Procedure

1. Initialize separate vectors for IDs, age values, gender assignments, and income numbers within R environment.
2. Utilize barplot() to visually capture age distribution variance.
3. Tabulate categories and leverage pie() function to display proportional splits.
4. Arrange attributes inside a cohesive R data.frame structural schema.
5. Move flat source data to Tableau environment to design sheets and stitch them into a unified dashboard layout.

RESULT:

The customer demographic attributes were effectively modeled and visualized through R graphical packages, followed by building a responsive dashboard in Tableau to support comprehensive exploratory data operations.

